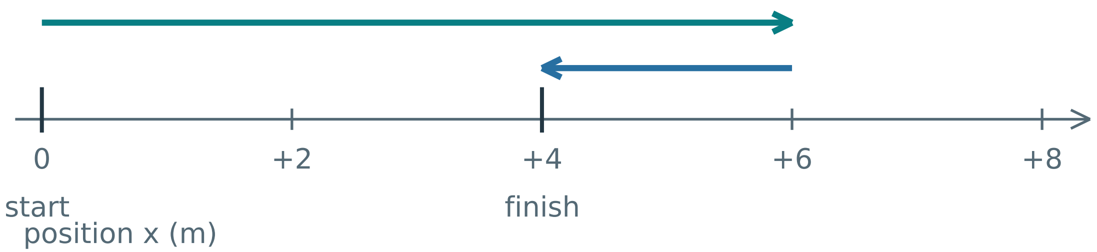
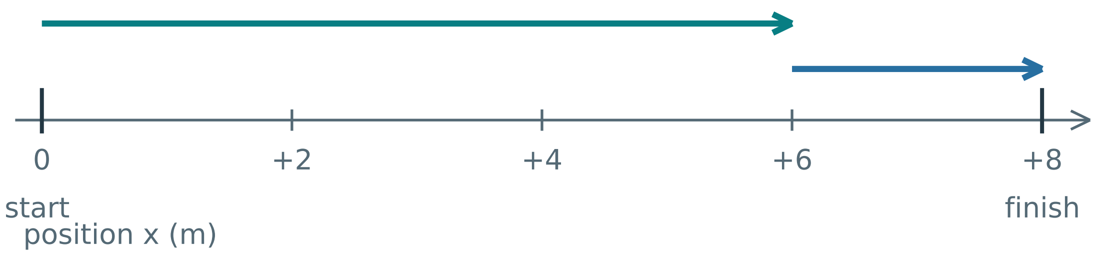
The same path length can end in different places

Both journeys start at 0 m and cover 8 m. Compare what changes.

A: 6 m right, then 2 m left



B: 6 m right, then 2 m right



In A, part of the route is retraced. In B, every part contributes to the same rightward change.

Compare the two totals

Both distances: 6 + 2 = 8 m.
Displacement A: +6 − 2 = +4 m.
Displacement B: +6 + 2 = +8 m.

Reconstruct the rule

$$\Delta x = x_f - x_i$$

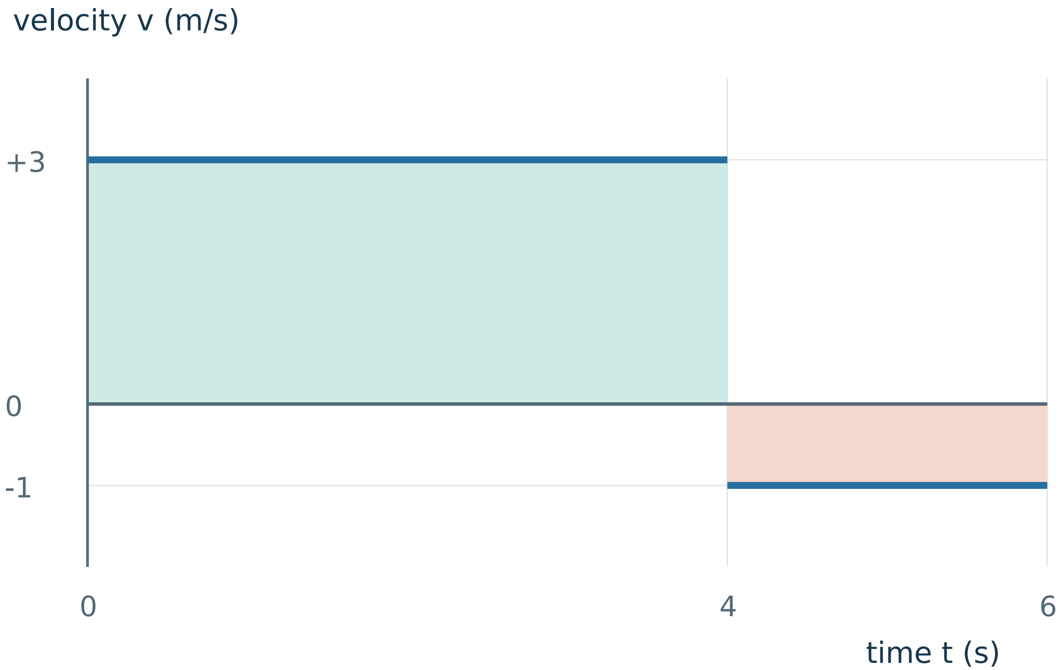
Δx means change in position; x_i is the initial position and x_f the final position.

Distance adds path lengths. The bars in $|\Delta x|$ mean size without the sign. This size cannot exceed the path length; equality holds with no reversal.

Changing the origin changes both coordinates equally, so their difference is unchanged. Reversing the positive axis changes the sign of Δx, but not the distance.

Read a graph as accumulated movement

Take right as positive. A constant velocity gives one rectangle of movement.



Areas are +12 m and −2 m. The second width is 6 − 4 = 2 s, not 6 s.

Build from a constant interval

$$\Delta x = v\,\Delta t$$

Velocity is signed movement per second; multiplying by elapsed time reconstructs the movement.

Thus graph height × width gives displacement. Units: (m/s) × s = m.

The abrupt switch at t = 4 s is an idealisation. It contributes no extra time or distance.

$$\Delta x = +12 - 2 = +10 \text{ m}$$

$$\text{Distance} = 12 + 2 = 14 \text{ m}$$

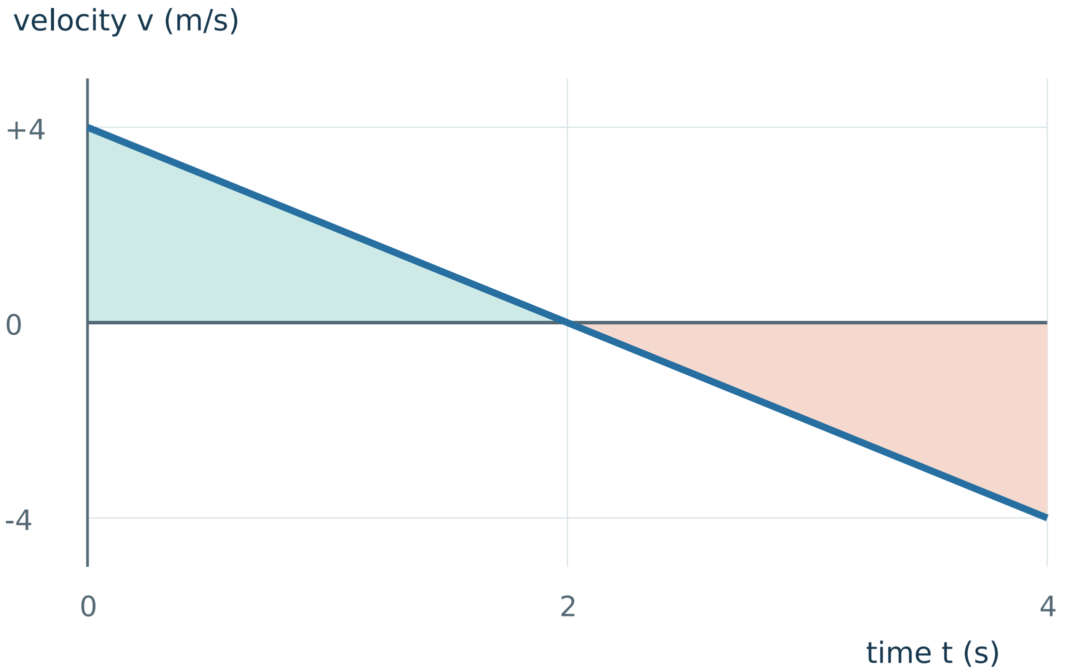
Final position still needs a start: $x_f = x_i + \Delta x$. A v–t graph alone does not provide x_i.

Keep graph height, slope and area separate: height is velocity; slope is its rate of change; signed area is displacement.

2 · SPLIT WHERE THE DIRECTION CHANGES

A straight segment can cross zero

Velocity falls linearly from +4 m/s at 0 s to −4 m/s at 4 s.



Each triangle has base 2 s and height size 4 m/s. The sign changes at 2 s.

Predict before calculating

The two equal area sizes cancel in displacement. Both still count as movement.

$$\Delta x = +\tfrac{1}{2}(2)(4) - \tfrac{1}{2}(2)(4)$$

$$= 0 \text{ m}$$

$$\text{Distance} = 4 + 4 = 8 \text{ m}$$

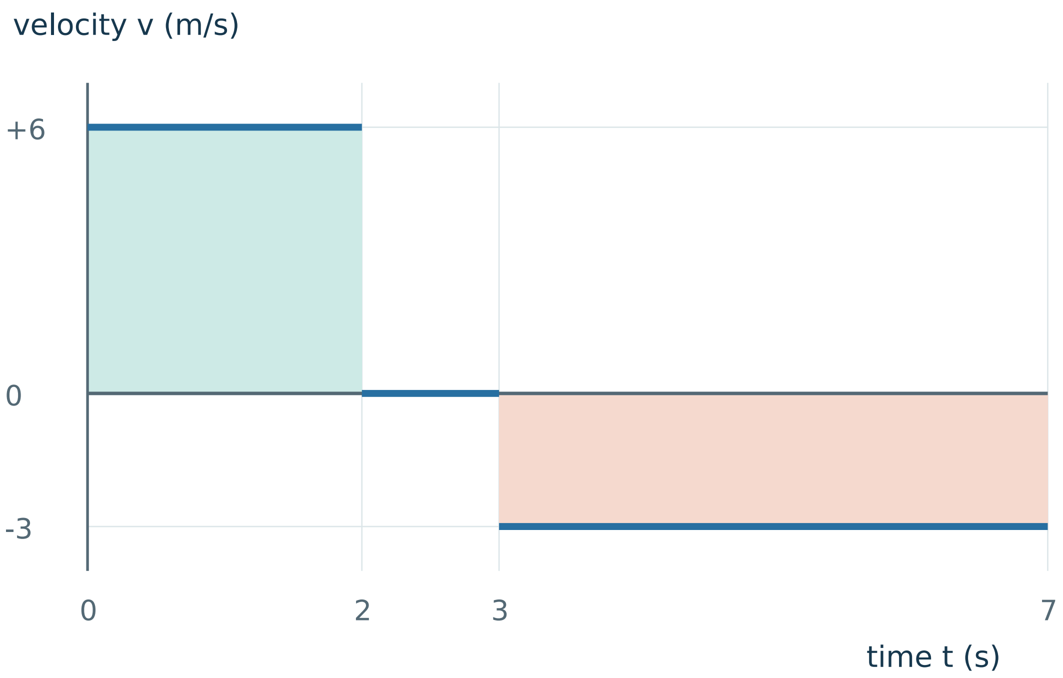
The ½ comes from triangular geometry. Straight graph segments let us calculate these areas exactly.

Zero displacement can hide substantial movement. Returning to the start does not imply continuous rest.

A zero value at an instant is not sufficient to prove a reversal. Check whether velocity actually changes sign on the two sides of that instant.

A stopped interval changes time, not path length

A sensor records +6 m/s for 2 s, rest for 1 s, then −3 m/s for 4 s.



Movement: 12 m right, no movement for 1 s, then 12 m left. The return ends at the start.

Treat each interval honestly

The changes are +12 m, 0 m and −12 m. Rest contributes zero area, even though time passes.

The vertical jumps at 2 s and 3 s idealise changes that take no time, so they add no area or distance.

$$\Delta x = +12 + 0 - 12 = 0 \text{ m}$$

$$\text{Distance} = 12 + 0 + 12 = 24 \text{ m}$$

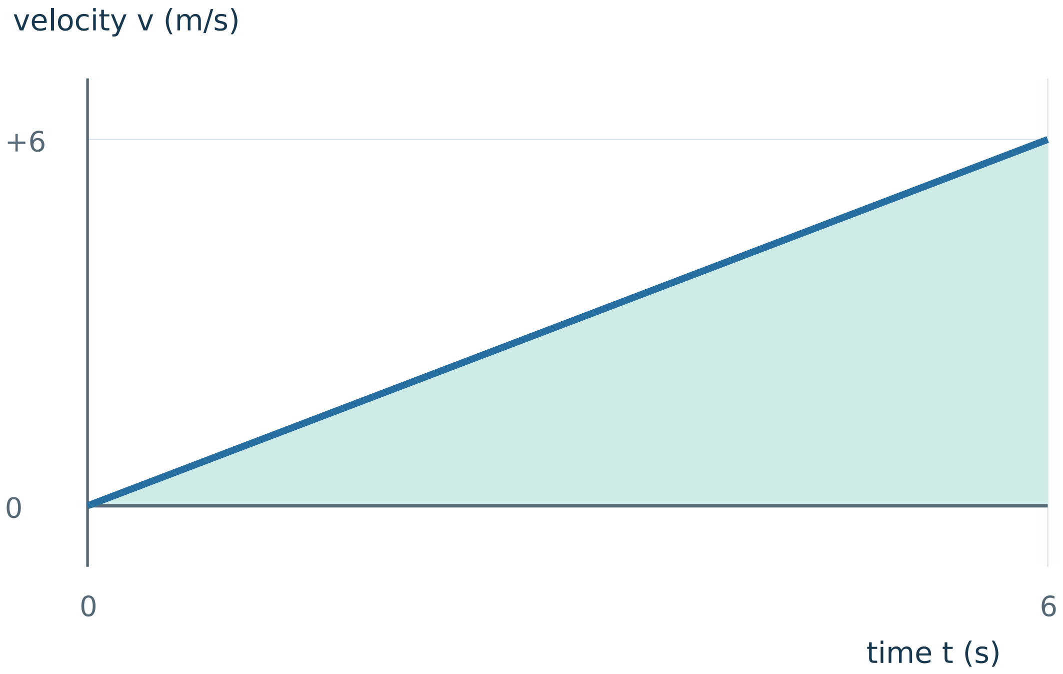
Change the starting position

Start at +9 m and finish at +9 m. The same areas still give zero displacement and 24 m distance.

Check a solution using both representations: a signed-area calculation and a right–stop–left motion sketch must tell the same story.

Why the final height can mislead you

Velocity rises linearly from 0 to +6 m/s over 6 s.



The actual area is half the rectangle formed by the final velocity and the full duration.

Test the tempting shortcut

Using 6 × 6 assumes the object travels at +6 m/s throughout. That contradicts the graph.

$$\Delta x = \tfrac{1}{2} \times 6 \times 6 = +18\ \text{m}$$

Distance is 18 m because there is no negative-velocity interval.

What if the line is curved?

Do not force it into a triangle. Signed area still represents displacement, but its numerical evaluation needs more information or an approximation.

For independent practice, choose your own diagram. A shorter explanation is acceptable only if the model, signs and physical check remain clear.

Try A7 · p. 10

Try A8 · p. 11

Appendix A · Core practice

Use the diagrams you need. A correct number needs a physical explanation.

A1 · Apply

Start at +1 m. Walk 5 m left, then 3 m right. Find the final position, distance and displacement.

Start → turning point → finish
Final position: ____
Distance: ____ Displacement: ____

Hints · p. 15 Solution · p. 17

A2 · Apply

An object moves from −2 m to +5 m without reversing. Find its distance and displacement. Explain the signs.

Sketch the line and explain why crossing zero is not a turn.

Hints · p. 15 Solution · p. 17

Appendix A · Core practice

Use the diagrams you need. A correct number needs a physical explanation.

A3 · Explain

Two students start at 0 m and finish at +4 m. Must they have travelled the same distance? Draw two journeys to support your answer.

Draw and label two routes. Keep the endpoints the same.

Hints · p. 15 Solution · p. 18

A4 · Transfer

A particle goes from −3 m to +5 m, then returns to +1 m. Find distance and displacement. A second observer adds 10 m to every coordinate. What changes?

Show the route or working that makes your answer clear.

Hints · p. 15 Solution · p. 18

Appendix A · Core practice

Use the diagrams you need. A correct number needs a physical explanation.

A5 · Compare

Journey A: +3 m/s for 4 s, then +2 m/s for 2 s. Journey B changes only the last velocity to −2 m/s. Find both distances and displacements. Explain what stays the same.

Show the route or working that makes your answer clear.

Hints · p. 16

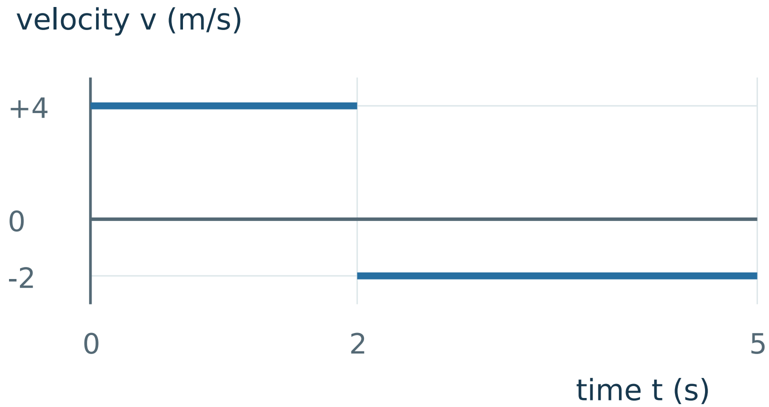
Solution · p. 19

Appendix A · Core practice

Use the diagrams you need. A correct number needs a physical explanation.

A6 · Apply

Use the graph to find displacement and distance from 0 s to 5 s.
Give a physical meaning for each area.



Hints · p. 16

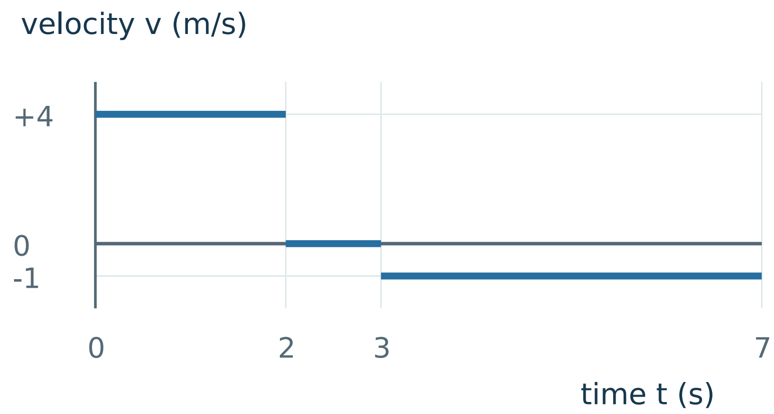
Solution · p. 19

Appendix A · Core practice

Use the diagrams you need. A correct number needs a physical explanation.

A7 · Transfer

A robot has the graph shown and starts at x = −5 m. Find its distance, displacement and final position after 7 s.



Hints · p. 16

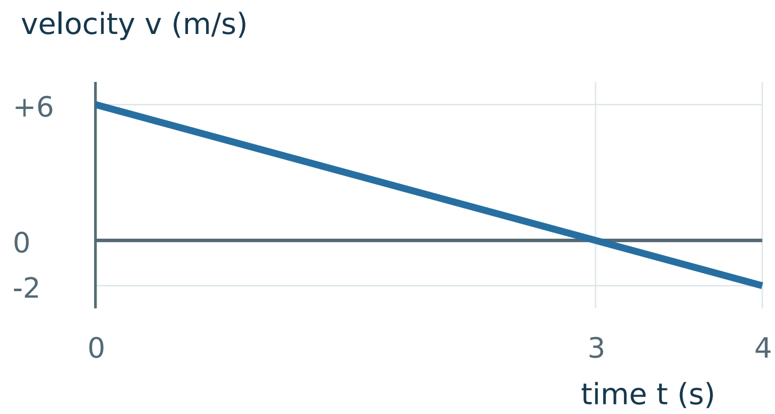
Solution · p. 20

Appendix A · Core practice

Use the diagrams you need. A correct number needs a physical explanation.

A8 · Transfer

The velocity follows the straight graph shown. Find distance and displacement over 4 s. At what time does the particle reverse direction? Justify this from the graph.



Hints · p. 16

Solution · p. 20

Mixed transfer · Concepts hidden

Work from the physical situation. Concept and task labels appear only after marking.

Question 1

Start at +1 m. Walk 5 m left, then 3 m right. Find the final position, distance and displacement.

Start → turning point → finish
Final position: ____
Distance: ____ Displacement: ____

Check later · p. 17

Question 2

An object moves from −2 m to +5 m without reversing. Find its distance and displacement. Explain the signs.

Sketch the line and explain why crossing zero is not a turn.

Check later · p. 17

Mixed transfer · Concepts hidden

Work from the physical situation. Concept and task labels appear only after marking.

Question 3

Journey A: +3 m/s for 4 s, then +2 m/s for 2 s. Journey B changes only the last velocity to −2 m/s. Find both distances and displacements. Explain what stays the same.

Show the route or working that makes your answer clear.

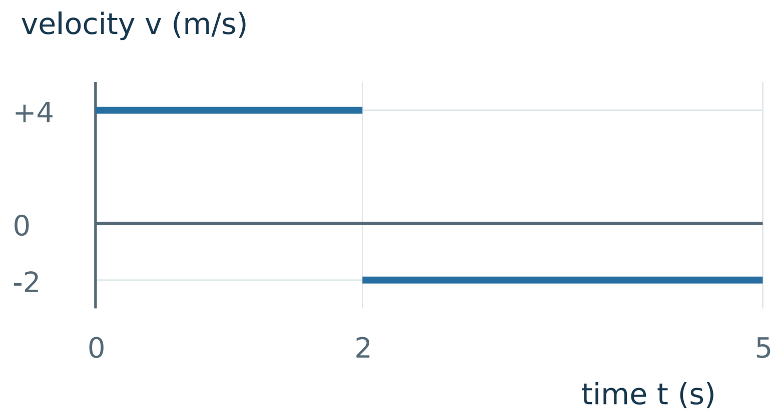
Check later · p. 19

Mixed transfer · Concepts hidden

Work from the physical situation. Concept and task labels appear only after marking.

Question 4

Use the graph to find displacement and distance from 0 s to 5 s.
Give a physical meaning for each area.



Check later · p. 19

Appendix B · Optional hints

Cover the rows below the hint you are reading.

A1

H1 Notice Keep the starting coordinate separate from the lengths walked.

H2 Model Mark the start, the turning point and the finish on a number line.

H3 Start The turning point is at $+1 - 5$ m. Find the finish next.

Return · p. 6

A2

H1 Notice Crossing the origin does not mean reversing direction.

H2 Model Draw -2, 0 and $+5$ in order along the line.

H3 Start The first part of the path, from -2 m to 0 m, is 2 m.

Return · p. 6

A3

H1 Notice The question fixes the endpoints, not the route.

H2 Model Draw one direct route and one route with a turn.

H3 Start For the turning route, choose a turning point beyond $+4$ m.

Return · p. 7

A4

H1 Notice The second observer changes the labels, not the physical journey.

H2 Model Draw the three positions, then label each with 10 m added.

H3 Start Compare $(1 + 10) - (-3 + 10)$ with $1 - (-3)$.

Return · p. 7

Appendix B · Optional hints

Cover the rows below the hint you are reading.

A5

H1 Notice Only the direction of the last interval changes.

H2 Model Sketch the two graphs with the same scales.

H3 Start Calculate the common first area: (+3 m/s) × (4 s).

Return · p. 8

A6

H1 Notice Read the duration of each horizontal segment.

H2 Model Separate the two rectangular regions at $t = 2$ s.

H3 Start For the first region, write $(+4) \times (2 - 0)$.

Return · p. 9

A7

H1 Notice The starting coordinate is separate from the displacement.

H2 Model Pair the graph regions with right, stopped and left motion arrows.

H3 Start The final region has duration $7 - 3$ s.

Return · p. 10

A8

H1 Notice Locate the zero crossing before calculating areas.

H2 Model Use the straight-line change to split the graph into two triangles.

H3 Start The velocity decreases by $8 \div 4$ m/s each second.

Return · p. 11

Appendix B · Full solutions

If a step surprised you, revisit the linked lesson and explain its picture.

QUESTION RECAP · A1 Start +1 m; 5 m left, then 3 m right.

WHY THIS WORKS. Position changes carry direction; path lengths do not.

METHOD. Track the walk step by step: start at +1 m, walk 5 m left to reach 1 − 5, then walk 3 m right to reach the final position, 1 − 5 + 3. Distance simply adds both leg lengths, 5 + 3, while displacement compares only the final position to the start, −1 − (+1).

ANSWER / CHECK. Final position −1 m; distance 8 m; displacement −2 m (left).

CONCEPT TO KEEP. The final coordinate is not automatically the displacement.

Return to A1 · p. 6 Lesson · p. 1

QUESTION RECAP · A2 Move from −2 m to +5 m with no reversal.

WHY THIS WORKS. The route goes 2 m to zero and another 5 m right.

METHOD. Distance adds the two legs of the actual path travelled: 2 m to reach zero, then 5 m more to +5 m, giving 2 + 5. Displacement instead compares only the two endpoints, final position minus initial position: +5 − (−2).

ANSWER / CHECK. Distance 7 m; displacement +7 m (right).

CONCEPT TO KEEP. Equal distance and displacement size require no reversal in this straight-line journey.

Return to A2 · p. 6 Lesson · p. 1

Appendix B · Full solutions

If a step surprised you, revisit the linked lesson and explain its picture.

QUESTION RECAP · A3 Same start 0 m and finish +4 m; compare possible distances.

WHY THIS WORKS. Endpoints fix displacement but leave the path undecided.
METHOD. One can go directly 0 → +4. Another can go 0 → +7 → +4. Their paths are 4 m and 7 + 3 m.
ANSWER / CHECK. No. These distances are 4 m and 10 m; both displacements are +4 m.
CONCEPT TO KEEP. A drawing can disprove an “always” claim.

Return to A3 · p. 7 Lesson · p. 1

QUESTION RECAP · A4 Route −3 → +5 → +1 m; then shift every coordinate by +10 m.

WHY THIS WORKS. Displacement is a coordinate difference, so a common shift cancels.
METHOD. Path lengths: 5 − (−3) = 8 m and 5 − 1 = 4 m. Δx = 1 − (−3) = +4 m. Shifted endpoints are +7 m and +11 m.
ANSWER / CHECK. Distance 12 m and displacement +4 m for both origins. Positions change.
CONCEPT TO KEEP. A reference origin changes position values, not a fixed journey.

Return to A4 · p. 7 Lesson · p. 1

Appendix B · Full solutions

If a step surprised you, revisit the linked lesson and explain its picture.

QUESTION RECAP · A5 A: +3 for 4 s, then +2 for 2 s. B: last velocity becomes −2 m/s.

WHY THIS WORKS. Reversing the second velocity changes the area sign, not its size.

METHOD. First area is +12 m in both. The last area is +4 m in A and −4 m in B. Add signs for displacement and sizes for distance.

ANSWER / CHECK. A: $\Delta x = +16$ m, distance 16 m. B: $\Delta x = +8$ m, distance 16 m.

CONCEPT TO KEEP. A sign change can alter the endpoint without altering total path.

Return to A5 · p. 8 Lesson · p. 2

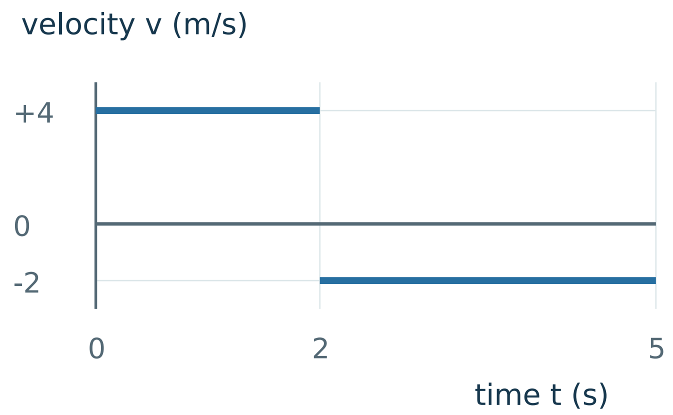
QUESTION RECAP · A6 Graph: +4 m/s from 0–2 s, then −2 m/s from 2–5 s.

WHY THIS WORKS. Graph widths are durations. The second width is $5 - 2 = 3$ s.

METHOD. Signed areas are $+4 \times 2 = +8$ m and $-2 \times 3 = -6$ m. For distance count both path lengths.

ANSWER / CHECK. Displacement +2 m; distance 14 m. The parts mean 8 m right and 6 m left.

CONCEPT TO KEEP. Read each width from its endpoints before multiplying.



Return to A6 · p. 9 Lesson · p. 2

Appendix B · Full solutions

If a step surprised you, revisit the linked lesson and explain its picture.

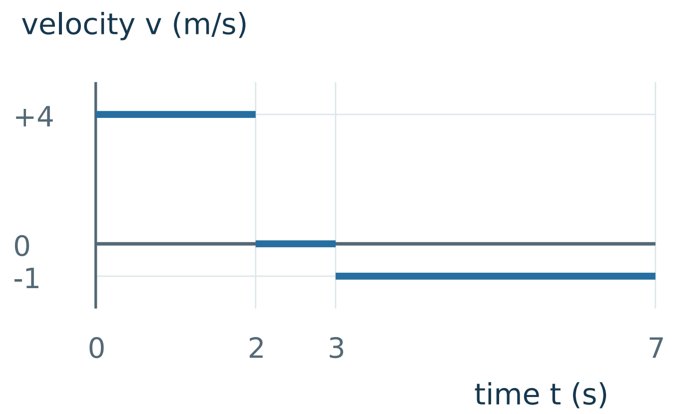
QUESTION RECAP · A7 Graph +4 m/s for 2 s, rest 1 s, then −1 m/s for 4 s; start −5 m.

WHY THIS WORKS. Rest adds no displacement. Final position is start plus signed change.

METHOD. Areas are +8 m, 0 m and −4 m. Add their sizes for distance and their signed values for displacement. Then add the change to −5 m.

ANSWER / CHECK. Distance 12 m; displacement +4 m; final position −1 m.

CONCEPT TO KEEP. A positive displacement can finish at a negative coordinate.



Return to A7 · p. 10 Lesson · p. 4

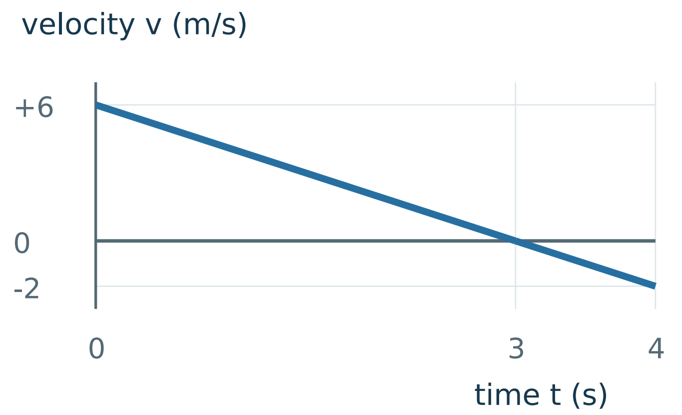
QUESTION RECAP · A8 Velocity falls linearly from +6 to −2 m/s over 4 s.

WHY THIS WORKS. Reversal occurs where velocity changes sign; split the signed area there.

METHOD. Velocity falls 8 m/s in 4 s. It reaches zero after 6 ÷ 2 = 3 s. Areas: +½ × 3 × 6 = +9 m; −½ × 1 × 2 = −1 m.

ANSWER / CHECK. Displacement +8 m; distance 10 m; reversal at 3 s.

CONCEPT TO KEEP. Equal endpoint signs are not the test; inspect the sign throughout the interval.



Return to A8 · p. 11 Lesson · p. 3

B80-MIXED-01

Mixed transfer · Mark and diagnose

Reveal these pages only after completing the mixed set.

Question 1 · Displacement

If this answer was weak, use the linked solution and its repair route before retrying.

Solution · p. 17

Question 2 · Distance

If this answer was weak, use the linked solution and its repair route before retrying.

Solution · p. 17

Question 3 · Signed v-t Area (Displacement)

If this answer was weak, use the linked solution and its repair route before retrying.

Solution · p. 19

Question 4 · v-t Area Magnitudes (Distance)

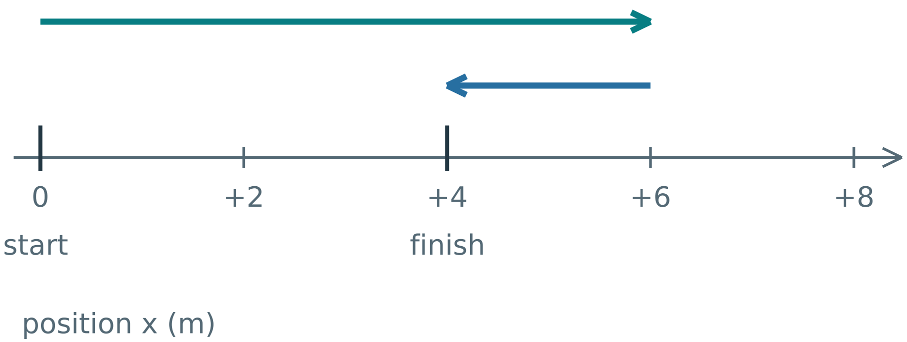
If this answer was weak, use the linked solution and its repair route before retrying.

Solution · p. 19

Appendix C · Keep the picture, rebuild the rule

Right is positive. Cover the words and explain each diagram aloud.

1 Distance and displacement



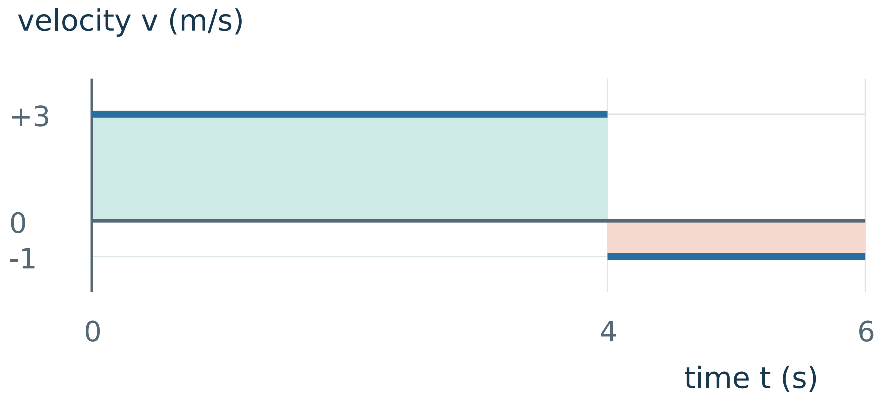
Distance: add every path length.
Displacement: the signed endpoint change.

$$\Delta x = x_f - x_i$$

Δx: displacement; x_i: initial position; x_f: final position. Here, distance is 8 m and Δx is +4 m.

Check: distance ≥ $|\Delta x|$. Bars mean size.
Rebuild: trace the path, then compare its endpoints.

2 Reading a velocity–time graph



Displacement: add signed areas.
Distance: add all the area sizes. Split at zero crossings.

Rectangle: v × duration

Triangle size: ½ × base × height.
Graph above: Δx = +10 m; distance 14 m.

Check: (m/s) × s = m. Use a rectangle for constant velocity. Use ½ × base × height only for a triangular region; other straight segments may need rectangle + triangle.